

Cell Design Specification — VBF-T6R1-DS-01

Case	t6_r1 — smartphone battery (ED ≥ 950 Wh/L, 4C no-plating, T_max ≤ 50 °C, SEI ≤ 500 nm @100 cyc, plateau ≥ 4.1 V)
Generation date	2026-08-25
Fidelity	DFN (highest available proxy; real_compute=false — no true DFT/MD endorsement)

1. Basic Specification

Item	Value	Source
Electrochemical system	LNMO high-voltage spinel cathode (4.7 V-class) / graphite anode; base parameter set `scripts/bda/simulators/data/LNMO.json` (Chen2020 + JSON overrides)	bridge/propose_r1.json; entry 0
Nominal capacity (parameter set)	4.5 Ah	parameter set `Nominal cell capacity [A.h]`
Simulation-verified capacity (1C, DFN)	6.198 Ah	cell/r8_d6_1c_dfn.json:capacity_ah
Voltage window	2.5 V – 4.7 V	parameter set Lower/Upper voltage cut-off
Voltage plateau (discharge midpoint)	4.1145 V	cell/r8_d6_energy_dfn.json:midpoint_voltage_v
Cell dimensions	65 mm (H) × 1580 mm (W) × 217.8 μm (electrode stack thickness); pouch shell thickness: Not provided (not modeled)	parameter set Electrode height/width; cell/r8_d6_energy_dfn.json:thickness_m
Electrode area	0.1027 m²	cell/r8_d6_energy_dfn.json:area_m2
Electrolyte formulation	EC/EMC + LiPF6 baseline; EC initial concentration 2270.5 mol/m³; EC diffusivity 1.0e-18 m²/s (SEI-limiting, diffusion-limited regime); electrolyte diffusivity 6.0e-10 m²/s constant (low-viscosity fast-charge formulation — caveat: replaces concentration-dependent Nyman2008 function, upper bound of reported carbonate ranges)	bridge/params_final.json
Cation transference number	0.2594	parameter set (dump)
Cooling	h = 400 W/m²K (total heat transfer coefficient)	bridge/params_final.json

2. Electrode and Separator

Layer	Thickness (μm)	Porosity	Active-material volume fraction	Particle radius (μm)	Density (kg/m³)	Source
Positive electrode (LNMO)	75.6	0.335	0.665	5.22	4400	parameter set dump
Negative electrode (graphite)	102.2	0.25	0.75	3.5	1657	parameter set dump + params_final.json
Separator	12	0.47	—	—	—	parameter set dump
Positive current collector (Al)	16	—	—	—	2700 (layer mass 0.0432 kg/m²)	parameter set dump; calc-energy
Negative current collector (Cu)	12	—	—	—	8960 (layer mass 0.10752 kg/m²)	parameter set dump; calc-energy

N/P ratio = 0.80. Formula (contract caliber): $N/P = (c_{\text{max,neg}} \times \epsilon_{\text{am,neg}} \times L_{\text{neg}}) \div (c_{\text{max,pos}} \times \epsilon_{\text{am,pos}} \times L_{\text{pos}}) = (33133 \times 0.75 \times 102.2\text{e-}6) \div (63104 \times 0.665 \times 75.6\text{e-}6) = 0.8005$.

Honest labeling correction: earlier rounds referred to the shipping anode thickness (102.2 μm) as "N/P 1.2" — that was a thickness-ratio shorthand vs the Chen2020 default negative thickness (85.2 μm), not a capacity-ratio N/P. The mechanical computation above supersedes it. *Parameter-set caveat:* LNMO.json carries positive capacity under key "Positive electrode maximum concentration [mol.m-3]", which is inert in PyBaMM 26.7 (current name: "Maximum concentration in positive electrode [mol.m-3]"), so the merged set uses Chen2020's positive $c_{\text{max}} = 63104 \text{ mol/m}^3$ — library data quirk, recorded.

3. Process Design Parameters

Parameter	Value	Formula	Notes
Positive areal density	221.2 g/m²	$75.6\text{e-}6 \times 0.665 \times 4400 \times 1000$	coating solid-phase mass
Negative areal density	127.0 g/m²	$102.2\text{e-}6 \times 0.75 \times 1657 \times 1000$	coating solid-phase mass

Positive compaction density	2.93 g/cm³	4400 × 0.665 ÷ 1000	divide by 1000
Negative compaction density	1.24 g/cm³	1657 × 0.75 ÷ 1000	divide by 1000
Electrolyte fill amount	6.27 g	pore volume 5.804 cm³ × 1.2 g/cm³ × 0.9 fill factor	electrolyte density = literature value 1.2 g/cm³ (annotated); pore volume = Σ(thickness × porosity × area)
Formation	0.1C CC to 4.7 V, 25 °C, 2 cycles	—	design-recommended value; actual production-line value requires tuning (annotated)

4. Mass Breakdown

Contract caliber (calc-energy): mass = Σ layer kg/m² × area; electrolyte excluded (parameter set lacks density).

Component	kg/m²	Mass (g)	Source
Positive electrode coating	0.221206	22.72	cell/r8_d6_energy_dfn.json:layer_kg_m2 × area_m2
Negative electrode coating	0.127009	13.04	same
Positive CC (Al)	0.043200	4.44	same
Negative CC (Cu)	0.107520	11.04	same
Separator	0.002525	0.26	same
Total (electrolyte excluded)	0.501460	51.50	cell/r8_d6_energy_dfn.json:mass_kg
Electrolyte (design estimate)	—	6.27	\$3 fill amount, literature density

5. Performance Verification (vs entry-0 criteria)

Metric	Value	Criterion	Verdict	Source
Volumetric energy density	1135.8 Wh/L	≥ 950 Wh/L	✓ (+19.6%)	cell/r8_d6_energy_dfn.json:energy_density_wh_l
Gravimetric energy density	493.3 Wh/kg	informational	—	cell/r8_d6_energy_dfn.json:energy_density_wh_kg
Voltage plateau (midpoint)	4.1145 V	≥ 4.1 V	✓ (+14.5 mV)	cell/r8_d6_energy_dfn.json:midpoint_voltage_v
4C charge max temperature	321.42 K (48.27 °C)	≤ 323.15 K	✓ (margin 1.73 K)	cell/r8_d6_4c_dfn.json:T_max_K
4C charge plating	min anode potential +10.96 mV	plated = false	✓ (margin 11.0 mV)	cell/r8_d6_4c_dfn.json:anode_potential_v_min
SEI thickness after 100 cycles	385.3 nm	≤ 500 nm	✓ (margin 114.7 nm)	cell/r8_d6_aging_dfn.json:sei_thickness_nm_end

All five contract criteria PASS at DFN fidelity from one params file (bridge/params_final.json).

6. Design Notes (which parameters were changed and why)

Change	Reason	Audit source
System switch: Chen2020 NMC811 → LNMO (4.7 V spinel)	Chen2020 midpoint 3.935 V < 4.1 V — material bottleneck proven by simulation	log R1 evaluate (systemLNMO)
Electrolyte: EC initial concentration 4541 → 2270.5 mol/m³; EC diffusivity 2e-18 → 1.0e-18 m²/s	SEI growth in EC-diffusion-limited regime ($L \propto \sqrt{c_{\text{EC}} \cdot D_{\text{EC}}}$); SEI 753 → 385 nm	log R4–R5 evaluate
Anode thickness 85.2 → 102.2 μm; negative particle radius → 3.5 μm	end-of-charge anode surface-saturation dip (plating)	log R3–R4 evaluate
Electrolyte diffusivity → 6.0e-10 m²/s constant (fast-charge formulation)	DFN through-plane depletion sag (midpoint 4.0974) and end-of-charge gradients (plating −19.3 mV); +14.5 mV midpoint / +30.3 mV anode margin	log R7–R8 evaluate
Cooling h 10 → 400 W/m²K	DFN heat load −6 W pushed T_max to 323.80 K; h=400 → 321.42 K	log R2, R7 evaluate
Radius 3.0 μm probe dropped	counterintuitive: smaller particles worsened DFN midpoint by ~13 mV (depletion profile)	log R7 evaluate (nr30)